

University of Naples Federico II
Web Technologies Course

RegexRiddle

Project documentation

Student

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Track

4.D, RegexRiddle

Professor

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Academic year

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Technology stack

Web platform for regular expression riddles: a challenge hides a regular expression, and players write one they believe behaves the same way.

Back end

Node.js 22	Runtime for the API, the seed and the test suite
Express 5	HTTP layer: routing, middleware and the REST endpoints
TypeScript	Language, on this side and the other
PostgreSQL 17	Relational store for the game data and the sessions
Prisma	Typed database client, with migrations generated from one schema
RE2	Linear-time regex engine; every user pattern runs through it
argon2	Password hashing, argon2id
zod	Strict validation of every request body and route parameter
express-session **connect-pg-simple**	Session cookies, with the sessions themselves in PostgreSQL
helmet	Protective HTTP headers
multer	Multipart parsing for the avatar upload
Docker Compose	Runs PostgreSQL for development and for the demo

Front end

Vue 3	Single-page application: Composition API, single-file components
Vite	Dev server with the `/api` proxy, and the production build
Vue Router	Client-side routing and the redirect back after a login
Pinia	Store for the session user and the accessibility preferences
TypeScript	Typed API client mirroring the REST contract
CSS custom properties	Theming, with no CSS framework
Atkinson Hyperlegible **OpenDyslexic**	Self-hosted typefaces, bundled so nothing loads from a CDN

Tests

Playwright	24 end-to-end tests against the running stack
tsx, node:assert	API suite, nine files against a live database